

CodeQuest — User Management Dashboard

A sample Angular application used as part of a technical interview for senior software developer candidates. The app is intentionally functional but contains a number of code-quality issues and bugs for the candidate to identify and fix.

Prerequisites

- **Node.js** ≥ 20
- **npm** ≥ 10
- **Angular CLI** ≥ 21

```
npm install -g @angular/cli
```

Installation

```
git clone <repository-url>
cd CodeQuest
npm install
```

Development server

```
ng serve
```

Open <http://localhost:4200>. The app auto-reloads on file changes.

Project structure

```
CodeQuest/
├── projects/
│   ├── code-quest-app/           # Main application
│   │   ├── src/app/
│   │   │   ├── dashboard/       # Dashboard page with stats
│   │   │   ├── app.ts           # Root component
│   │   │   ├── app.routes.ts    # Routing
│   │   │   └── app.config.ts     # App providers (HTTP, animations)
│   │   └── shared-lib/          # Shared models, pipes, components, mock backend
│   │       ├── src/lib/
│   │       │   ├── models/       # User model & types
│   │       │   ├── interceptors/ # Mock API interceptor (replaces real backend)
│   │       │   ├── components/   # MatDialog, LoadingSpinner
│   │       │   └── pipes/        # RoleBadgeColor, StatusLabel
```

```
├── users-lib/                # User feature library
│   └── src/lib/
│       ├── services/        # UserService (CRUD via HttpClient)
│       ├── user-list/       # User table with sort, filter, pagination
│       ├── user-form-dialog/ # Create / edit dialog (Reactive Forms)
│       └── user-detail/     # User detail view
└── tsconfig.json            # Path aliases: shared-lib, users-lib → source
```

Mock backend

There is no real backend. HTTP calls are intercepted by `mockApiInterceptor` (in `shared-lib`) which handles all `/api/users` endpoints in-memory with 20 seeded users.

This workspace runs **zoneless** (via `provideZonelessChangeDetection()` in `app.config.ts`), so the mock API returns **GET** responses synchronously to keep the UI reactive without Zone.js.

Running unit tests

```
ng test
```

Tests run with `Vitest` + `jsdom`.

Building

```
ng build
```

Build artifacts are placed in `dist/`.

Tech stack

Framework	Angular 21 (standalone components)
UI library	Angular Material + CDK
Forms	Reactive Forms
HTTP	<code>HttpClient</code> with functional interceptor
Tests	Vitest 4
Build	<code>@angular/build</code> (esbuild)

```
ng e2e
```

Angular CLI does not come with an end-to-end testing framework by default. You can choose one that suits your needs.

Additional Resources

For more information on using the Angular CLI, including detailed command references, visit the [Angular CLI Overview and Command Reference](#) page.